UNIT–V: CORRELATION & REGRESSION

COMPLETE QUESTION BANK (FINAL)

PART–A: THEORY QUESTIONS

Q1.Define correlation. Explain different types of correlation.

Q2.State the principles of least squares. Write the normal equations for fitting a straight line.

Q3.List and explain the properties of Karl Pearson's coefficient of correlation.

Q4.Define rank correlation coefficient. State its properties.

PART–B: KARL PEARSON CORRELATION

Q5.Calculate the coefficient of correlation from the following data:

X	12	9	8	10	11	13	7
Y	14	8	6	9	11	12	3

Q6.Calculate the coefficient of correlation from the following data:

X	15	18	20	24	30	35	40	50
Y	85	93	95	105	120	130	150	160

Q7.Calculate the coefficient of correlation from the following data:

X	10	15	12	17	13	16	24	14	22	20
Y	30	42	45	46	33	34	40	35	39	38

Q8.Calculate the coefficient of correlation from the following data:

X	9	8	7	6	5	4	3	2	1
Y	15	16	14	13	11	12	10	8	9

Q9.Calculate the coefficient of correlation between x and y:

X	1	2	3	4	5	6	7
Y	2	4	5	3	8	6	7

Q10.Calculate the coefficient of correlation between x and y:

X	1	2	3	4	5	6	7	8	9

Y	12	11	13	15	14	17	16	19	18

Q11.Calculate the coefficient of correlation from the following data:

X	**10**	**12**	**18**	**24**	**23**	**27**
Y	13	18	12	25	30	10

PART–C: RANK CORRELATION

Q12.Calculate the rank correlation coefficient from the following data:

X	**9**	**8**	**7**	**6**	**5**	**5**	**4**
Y	15	16	14	13	12	12	11

Q13. Calculate the rank correlation coefficient from the following data:

X	**3**	**5**	**8**	**4**	**7**	**7**	**10**
Y	6	4	9	8	1	2	3

Q14. Calculate the rank correlation coefficient from the following data:

X	**4**	**3**	**8**	**4**	**8**	**8**	**10**
Y	6	4	6	4	1	2	2

Q15.From the following data, calculate the rank correlation coefficient after adjusting for tied ranks:

X	**48**	**33**	**40**	**9**	**16**	**16**	**65**	**24**	**16**	**57**
Y	13	13	24	6	15	4	20	9	6	19

Q16.The following are the ranks obtained by 10 students in two subjects. Find the rank correlation coefficient:

Statistics	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**
Mathematics	2	4	1	5	3	9	7	10	6	8

PART–D: REGRESSION

Q17.

The equations of two regression lines are:

$7x - 16y + 9 = 0$

$5y - 4x - 3 = 0$

Find:
(a) Coefficient of correlation
(b) Means of x and y

Q18.Obtain the equations of two regression lines for the following data. Also estimate x when y = 70:

X	65	66	67	67	68	69	70	72
Y	67	68	65	68	72	72	69	71

Q19. From the following data, find the equations of the two regression lines:

X	1	2	3	4	5
Y	15	25	35	45	55

Q20.The following data relates to 450 students:

Mean of X = 40
Mean of Y = 8
Standard deviation of X = 12
Variance of Y = 256
Σ(xy) = 42075

Find:
(a) Regression equations
(b) Estimate Y when X = 50

PART–E: STRAIGHT LINE (LEAST SQUARES)

Q21.Fit a straight line to the following data:

X	1	2	3	4	5
Y	14	27	40	55	68

Q22.Fit a least squares straight line for the following data:

X	20	25	30	35	40	45	50
Y	0.18	0.37	0.35	0.78	0.56	0.75	1.18

Q23.Fit a straight line to the following data:

X	1	3	5	7	9
Y	1.5	2.8	4.0	4.7	6.0

Q24.Determine the equation of the straight line which best fits the data:

X	10	12	13	16	17	20	25
Y	10	22	24	27	29	33	37

Q25.Fit a straight line of the form $y = ax + b$ for the following data:

X	1	2	3	4	5
Y	4	3	6	7	11

PART–F: CURVE FITTING

Q26.Fit an exponential curve of the form $y = ae^{bx}$ for the following data:

X	1	2	3	4	5	6	7	8
Y	1.0	1.2	1.8	2.5	3.6	4.7	6.6	9.1

Q27.Fit an exponential curve $y = ae^{bx}$ for the following data:

X	1	2	3	4	5
Y	4	7	10	15	25

Q28.Fit a curve of the form $y = a + bx + cx^2$ for the following data:

X	0	1	2	3	4
Y	1	1.8	1.3	2.5	6.3

Q29.Find the curve of best fit using least squares method for the following data:

X	1	5	7	9	12
Y	10	15	12	15	21

Q30.Fit a suitable curve for the following data:

X	1	2	3	4	5	6
Y	151	100	61	50	20	8